

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

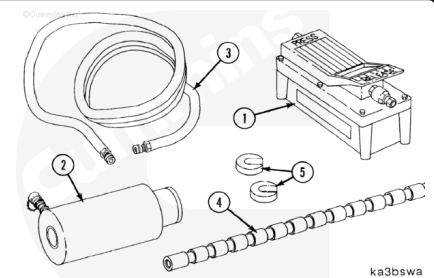
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the camshafts. Refer to Procedure 001-008 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-008-tr.html>).

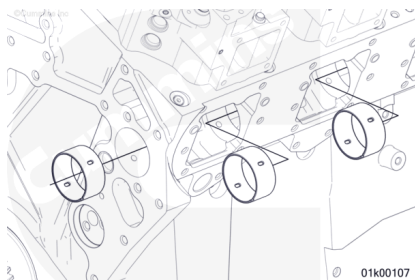
Remove

Use hydraulic actuator kit, Part Number 3823621, and camshaft bushing installation/removal kit, Part Number 3162257, to remove the camshaft bushings.

1. Hydraulic pump
2. Hydraulic/air cylinder
3. Hydraulic hose
4. Drive bar
5. Stop collars.



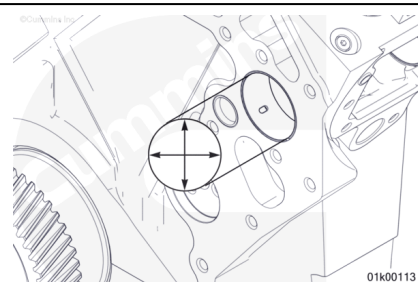
Remove the camshaft bushings.



Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



Use solvent to clean the camshaft bores in the block.

Use an emery cloth to remove any burrs and sharp edges from the bore.

Inspect the amount of copper exposure on the camshaft bushing inside diameter. If it extends more than 180 degrees around the bushing, the bushing **must** be replaced.

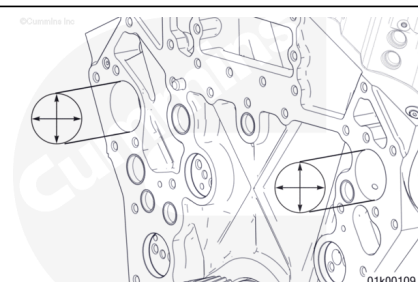
Measure the inside diameter of the bushing.

Camshaft Bushing Inside Diameter in Block		
mm		in
65.00	MIN	2.559
65.15	MAX	2.565

Note : If the bushing is not within specifications, the bushing **must** be replaced.

When bushings are replaced, the bushing bores in the block **must** be measured.

Camshaft Bushing Bores in the Block		
mm		in
69.01	MIN	2.717



Camshaft Bushing Bores in the Block

mm		in
69.04	MAX	2.718

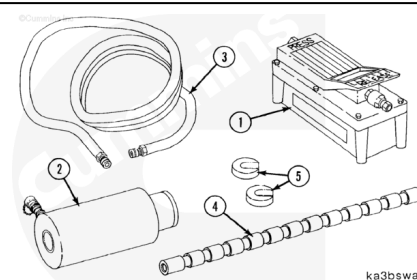
Note : If the camshaft bushing bore is not within specifications, the cylinder block must be replaced. No oversized camshaft bushings are available. Refer to Procedure 001-026 in Section 1.

(/qs3/pubsys2/xml/en/procedures/57/57-001-026.html)

Install

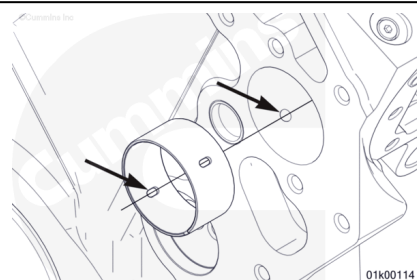
Use hydraulic actuator kit, Part Number 3823621, and camshaft bushing installation/removal kit, Part Number 3162257, to install the camshaft bushings.

1. Hydraulic pump
2. Hydraulic/air cylinder
3. Hydraulic hose
4. Drive bar
5. Stop collars.



Note : The oil holes in the bushing are off-center axially.

Align the oil holes in the bushing with the oil drillings in the block. There are two holes in each bushing. Both holes **must** be aligned. The end of the bushing closest to the holes is installed toward the front of the engine on the left bank, and toward the rear of the engine on the right bank.

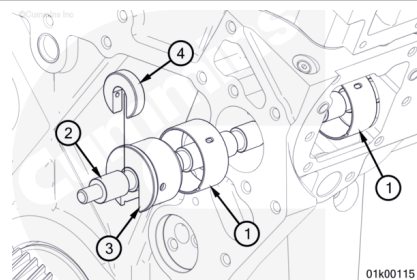


Install bushings in the bore through the front left bank and through the rear right bank of the cylinder block.

Slide each camshaft bushing (1) into its respective camshaft access port.

Insert and slide the drive bar (2) through each camshaft bushing and camshaft bore.

Install camshaft bushing replacer (3) on drive bar with smaller end of replacer toward hydraulic cylinder.

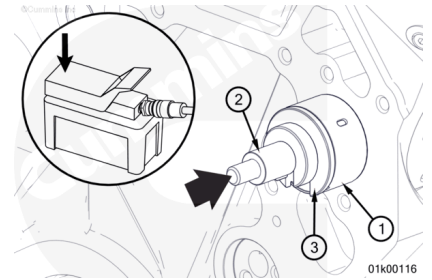


Slide bushing onto replacer. Slide this assembly toward camshaft bore.

Install stop collar (4) onto drive bar next to replacer as shown.

⚠ CAUTION ⚠

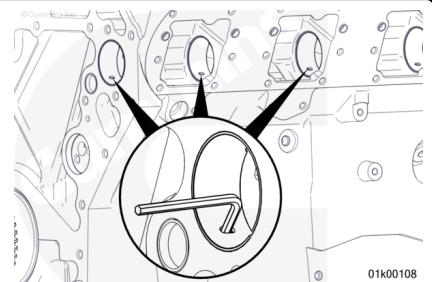
Confirm that the front bushing is installed below the surface of the block; otherwise damage to the component can occur.



Use hydraulic driver assembly and drive bar (2) to pull the camshaft bushing (1) and replacer (3) toward the rear of the cylinder block. Hydraulic cylinder pulls the camshaft bushing (2) into the camshaft bore.

Note : Camshaft bushing must be even with surface or slightly below with elongated oil holes aligned to oil holes in the cylinder block.

Use a 4.5-mm [11/64 in] Allen wrench or probe. Check the alignment of the bushing oil holes.



Finishing Steps

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⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Install the camshafts. Refer to Procedure 001-008 in Section 1.
(/qs3/pubsys2/xml/en/procedures/57/57-001-008-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine to 70°C [158°F] coolant temperature. Check for leaks.

Last Modified: 02-Nov-2017
